Calc II Section Notes

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Chapter 1

Integrals

Week 1 (8/31-9/3)

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Problem 1.1. Approximate the area under $f(x) = x^2$ on $[0, 1]$ using Riemann sums with $N = 2, 3$ and some choice of sample points. Then think about the limit as $N \to \infty$. Hint:

$$\sum_{i=1}^{N} i = 1 + 2 + \cdots + N = \frac{N(N+1)}{2}, \quad \sum_{i=1}^{N} i^2 = 1 + 4 + 9 + \cdots + N^2 = \frac{N(N+1)(2N+1)}{6}$$

Proof. We will use left endpoints. When $N = 2$, $x_1 = 0, x_2 = 1/2$. Thus

$$R_2 = 0 + \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$$

When $N = 3$, $x_1 = 0, x_2 = 1/3, x_3 = 2/3$. Then

$$R_3 = 0 + \frac{1}{9} \cdot \frac{1}{3} + \frac{4}{9} \cdot \frac{1}{3} = \frac{5}{27}$$

Now we compute when $N \to \infty$

$$R = \lim_{N \to \infty} \sum_{i=0}^{N-1} \left(\frac{i}{N}\right)^2 \frac{1}{N} = \frac{1}{6}$$

□

Problem 1.2. Compute and interpret the limit

$$\lim_{N \to \infty} \sum_{i=1}^{N} \left(\frac{i}{N^2} - \frac{i^2}{N^3}\right)$$

Proof. We first compute it:

$$\sum_{i=1}^{N}\left(\frac{i}{N^2}-\frac{i^2}{N^3}\right)=\frac{N(N+1)}{2N^2}-\frac{N(N+1)(2N+1)}{6N^3} \tag{1.1}$$

Letting $N\to\infty$,

$$\lim_{N\to\infty}\sum_{i=1}^{N}\left(\frac{i}{N^2}-\frac{i^2}{N^3}\right)=\frac{1}{2}-\frac{1}{3}=\frac{1}{6}$$

Now to interpret it, we can write it as a Riemann sum:

$$\sum_{i=1}^{N}\left(\frac{i}{N^2}-\frac{i^2}{N^3}\right)=\sum_{i=1}^{N}\frac{1}{N}\left(\frac{i}{N}-\frac{i^2}{N^2}\right)$$

Then $\Delta x=\frac{1}{N}$, and $x_i=\frac{i}{N}$. Letting $N\to\infty$, this is the Riemann sum of the following

$$\int_0^1 (x-x^2)\,dx$$

□

Problem 1.3. Compute

$$\int_{-1}^{5}(1+3x)\,dx$$

using Riemann sums.

Proof. We divide the interval $[-1,5]$ into N intervals, then $\Delta x=6/N$, and if we use right endpoints, the ith sample point is $x_i=-1+6i/N$. Then

$$\sum_{i=1}^{N}f(x_i)\Delta x=\sum_{i=1}^{N}(-2+18i/N)\frac{6}{N}$$

letting $N\to\infty$, the answer is 42. □

Week 2 (9/7-9/11)

Problem 1.4. Compute the following integral:

$$\int_1^3 \frac{\ln x}{x(1+\ln^2 x)}dx$$

Proof. Let $u = 1 + \ln^2 x$, then $du = \frac{2\ln x}{x}dx$, and

$$\int_1^3 \frac{\ln x}{x(1+\ln^2 x)}dx = \int_1^{1+\ln^2 3} \frac{1}{2u}du = \frac{1}{2}\ln(1+\ln^2 3)$$

□

Problem 1.5. Compute the following integral:

$$\int_0^{1/2} \frac{x}{\sqrt{1-x^4}}dx$$

Proof. Let $u = x^2, du = 2xdx$, then

$$\int_0^{1/2} \frac{x}{\sqrt{1-x^4}}dx = \int_0^{1/4} \frac{1}{2\sqrt{1-u^2}}du = \frac{1}{2}\left(\arcsin 1/4 - \arcsin 0\right)$$

□

Problem 1.6. Compute the following integral:

$$\int_1^2 \frac{e^x}{(1+e^x)\ln(1+e^x)}dx$$

Proof. Let $u = \ln(1+e^x)$, then $du = \frac{e^x}{1+e^x}dx$, thus

$$\int_1^2 \frac{e^x}{(1+e^x)\ln(1+e^x)}dx = \int_{\ln(1+e)}^{\ln(1+e^2)} \frac{1}{u}du = \ln\left(\frac{\ln(1+e^2)}{\ln(1+e)}\right)$$

□

Chapter 2

Applications of Integrals

Week 3 (9/14-9/18)

Problem 2.1. Compute the area enclosed by the following curves:

1. $$y = \frac{1}{x}, y = \frac{1}{x^2}, x = 2$$
2. $$x = 1 - y^2, x = y^2 - 1$$
3. $$x = \frac{\pi}{2}, x = \pi, y = \sin x, y = x$$
4. $$y = \sqrt[3]{2x}, y = \frac{1}{8}x^2, x = 0, x = 6$$

Proof. 1.
$$\int_1^2 \left(\frac{1}{x} - \frac{1}{x^2}\right) dx$$

2.
$$\int_{-1}^1 (1 - y^2) - (y^2 - 1) dy$$

3.
$$\int_{\pi/2}^{\pi} (x - \sin x)\, dx$$

4.
$$\int_0^4 \left(\sqrt[3]{2x} - x^2/8\right) dx + \int_4^6 \left(x^2/8 - \sqrt[3]{2x}\right) dx$$

□

Problem 2.2. Write down the integrals for the following solids of revolution using the washer method. DO NOT SOLVE.

1. $y = x, y = x^3, x \geq 0$, rotated about the x-axis.
2. $y^2 = x, x^2 = y$, rotated about $y = 1$.
3. $y^2 = x, x^2 = y$, rotated about the y-axis.
4. $x = 0, y = x^2, y = -2x + 3$ around the y-axis.

Proof.

1. The cross-sectional area in the x-direction is $\pi(x^2 - (x^3)^2)$, thus
$$S = \int_0^1 \pi(x^2 - x^6)dx$$
2. The cross-sectional area along the x-axis is
$$A = \pi\left[(1 - x^2)^2 - (1 - \sqrt{x})^2\right]$$
Thus the solid volume is
$$S = \int_0^1 \pi\left[(1 - x^2)^2 - (1 - \sqrt{x})^2\right] dx$$
3. The cross-sectional area along the y-axis is
$$\pi\left[(\sqrt{y})^2 - (y^2)^2\right] dy$$
Thus the solid volume is
$$S = \int_0^1 \pi\left[(\sqrt{y})^2 - (y^2)^2\right] dy$$
4. This needs to be separated into two terms:
$$S = \int_0^1 \pi(\sqrt{y})^2 dy + \int_1^3 \pi\left(\frac{(3-y)}{2}\right)^2 dy$$

□

Chapter 3

Techniques of Integration

Week 4 (9/21-9/25)

Problem 3.1. Compute

$$\int (\ln x)^2 dx$$

Proof. We do integration by parts twice:

$$\begin{cases} u = (\ln x)^2, dv = dx \\ du = \frac{2\ln x}{x} dx, v = x \end{cases}$$

Thus

$$\begin{aligned} \int (\ln x)^2 dx &= x(\ln x)^2 - \int 2\ln x dx \\ &= x(\ln x)^2 - 2\left(x\ln x - \int 1 dx\right) \\ &= x(\ln x)^2 - 2x\ln x + 2x + C \end{aligned}$$

□

Problem 3.2. Recall

$$\int e^x \sin x dx = \frac{1}{2} e^x (\sin x - \cos x) + C$$

Compute

$$\int e^{2\theta} \sin(3\theta) d\theta$$

Proof. Let $f' = e^x, g = \sin x$, then

$$\begin{aligned} \int e^x \sin x dx &= e^x \sin x - \int e^x \cos x dx \\ &= -\left(e^x \cos x + \int e^x \sin x\right) \qquad \text{(IBP twice)} \end{aligned}$$

This gives the implies result. Equivalently, we let $u = \sin x, du = \cos x dx, dv = e^x dx, v = e^x$. Now we compute

$$\begin{aligned}\int e^{2\theta}\sin(3\theta)d\theta &= \frac{1}{2}e^{2\theta}\sin(3\theta) - \frac{3}{2}\int e^{2\theta}\cos(3\theta)d\theta \\ &= \frac{1}{2}e^{2\theta}\sin(3\theta) - \frac{3}{2}\left(\frac{1}{2}e^{2\theta}\cos(3\theta) + \frac{3}{2}\int e^{2\theta}\sin(3\theta)d\theta\right)\end{aligned}$$

which implies

$$\int e^{2\theta}\sin 3\theta d\theta = \frac{4}{13}\left(\frac{1}{2}e^{2\theta}\sin 3\theta - \frac{3}{4}e^{2\theta}\cos 3\theta\right) + C$$

□

Problem 3.3. Compute

$$\int_0^1 (x^2+1)e^{-x}dx$$

Proof.

$$\begin{aligned}\int_0^1 (x^2+1)e^{-x}dx &= -e^{-x}(x^2+1)\Big|_0^1 + \int_0^1 e^{-x}2x dx \\ &= (1-2e^{-1}) - e^{-x}2x\Big|_0^1 + \int_0^1 e^{-x}2dx \\ &= 1 - 4e^{-1} - 2e^{-x}\Big|_0^1 \\ &= 3 - 6e^{-1}\end{aligned}$$

□

Week 4 (9/28-10/2)

Section 7.2 and 7.3

Proposition 3.1. For $\sin, \cos$:

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x) \qquad \text{and} \qquad \cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

$$2 \sin x \cos x = \sin 2x$$

For example, $\int \sin^4 x$.

Proposition 3.2. For $\tan, \sec$:

$$\tan^2 + 1 = \sec^2 x \qquad \text{and} \qquad \sec^2 x = \tan^2 x + 1$$

$$\int \tan x \, dx = \ln|\sec x| + C \qquad \text{and} \qquad \int \sec x \, dx = \ln|\sec x + \tan x| + C$$

For example, $\int \tan^3 x$.

Proposition 3.3.

(a) $\sin A \cos B = \frac{1}{2}[\sin(A - B) + \sin(A + B)]$

(b) $\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$

(c) $\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$

Problem 3.4. $\int \tan^2 x \, dx$

Proof.

$$\begin{aligned} \int \tan^2 x \, dx &= \int (\sec^2 x - 1) dx \\ &= \tan x - x + C \end{aligned}$$

□

Problem 3.5. $\int x \sin^2(x^2) \, dx$

Proof. First let $u = x^2$, then

$$\begin{aligned} \int x \sin^2(x^2) \, dx &= \frac{1}{2} \int \sin^2 u du \\ &= \frac{1}{4} \int 1 - \cos 2u du \\ &= \frac{x^2}{4} - \frac{\sin(2x^2)}{8} + C \end{aligned}$$

□

Problem 3.6. $\int \tan^3 x \sec x \, dx$

Proof. The power of $\tan$ is odd, so we want to save a factor of $\tan x \sec x$:

$$\begin{aligned}\int \tan^3 x \sec x \, dx &= \int \tan^2(\tan x \sec x) dx \\ &= \int (1 - \sec^2 x)(\tan x \sec x) dx \\ &= \int (1 - u^2) du\end{aligned}$$

□

Problem 3.7. $\int \sin 3x \sin 6x \, dx$

Proof.

$$\begin{aligned}\int \sin 3x \sin 6x \, dx &= \frac{1}{2}(\cos(3x) - \cos 9x) \\ &= \frac{1}{6}\sin 3x - \frac{1}{18}\sin 9x + C\end{aligned}$$

□

Problem 3.8. $\int \frac{x}{\sqrt{x^2 - 7}} \, dx$

Proof. Let $x = \sqrt{7}\sec\theta$, then

$$\begin{aligned}\int \frac{x}{\sqrt{x^2 - 7}} \, dx &= \int \frac{\sqrt{7}\sec\theta}{\sqrt{7}\tan\theta} \sqrt{7}\sec\theta\tan\theta d\theta \\ &= \int \sqrt{7}\sec^2 d\theta \\ &= \sqrt{7}\tan\theta + C \\ &= \sqrt{x^2 - 7} + C\end{aligned}$$

□

Problem 3.9. $\int_0^1 \sqrt{x^2 + 1} \, dx$

Proof. We let $x = \tan\theta$, then $dx = \sec^2\theta d\theta$, thus

$$\int_0^1 \sqrt{x^2 + 1} \, dx = \int_0^{\frac{\pi}{4}} \sec\theta \cdot \sec^2\theta d\theta$$

then one would use integraion by parts $u = \sec\theta, dv = \sec^2\theta d\theta$, to find

$$\begin{aligned}\int_0^1 \sqrt{x^2+1}\,dx &= \sec\theta\tan\theta - \int \sec\theta\tan^2\theta d\theta \\ &= \sec\theta\tan\theta - \int \sec^3\theta d\theta + \int \sec\theta d\theta\end{aligned}$$

which implies

$$\int_0^1 \sqrt{x^2+1}\,dx = \frac{1}{2}\left(\sec\theta\tan\theta + \ln|\sec\theta+\tan\theta|\right)\Big|_0^{\pi/4}$$

□

Week 5 (10/5-10/9)

Section 7.4 Partial Fractions.

Problem 3.10. Evalute the following integral:

$$\int \frac{1}{\sqrt{x}-\sqrt[3]{x}}dx$$

Hint: let $u = x^{\frac{1}{6}}$.

Proof.

$$\begin{aligned}\int \frac{1}{\sqrt{x}-\sqrt[3]{x}}dx &= \int \frac{6u^5}{u^3-u^2}du \\ &= 6\int \frac{u^3}{u-1}du \\ &= 6\int u^2+u+1+\frac{1}{u-1}du \\ &= 6\left(\frac{u^3}{3}+\frac{u^2}{2}+u+\ln|u-1|\right)+C\end{aligned}$$

then one can substitute $u = x^{\frac{1}{6}}$ in. □

Problem 3.11. Evalute the following integral:

$$\int x\tan^{-1}x dx$$

Proof.

$$\begin{aligned}\int x\tan^{-1}x dx &= \frac{x^2}{2}\tan^{-1}x - \frac{1}{2}\int \frac{x^2}{x^2+1}dx \\ &= \frac{1}{2}x^2\tan^{-1}x - \frac{1}{2}x + \frac{1}{2}\int \frac{1}{x^2+1}dx \\ &= \frac{1}{2}x^2\tan^{-1}x - \frac{1}{2}x + \frac{1}{2}\tan^{-1}x + C\end{aligned}$$

for some constant C. □

Problem 3.12. Evalute the following integral:

$$\int \frac{\sec^2 t}{\tan^2 t+3\tan t+2}dt$$

Proof. Let $u = \tan t$,

$$\begin{aligned}\int \frac{\sec^2 t}{\tan^2 t+3\tan t+2}dt &= \int \frac{1}{u^2+3u+2}du \\ &= \int -\frac{1}{u+2}du + \int \frac{1}{u+1}du \\ &= \ln\left|\frac{u+1}{u+2}\right| + C\end{aligned}$$

for some constant C. □

Ways to solve integrals so far: u-substitution, integration by parts, trig substitution, partial fractions, improper integrals.

Week 6 (10/12-10/16)

Problem 3.13. Compute the following improper integral:

$$\int_1^\infty \frac{\ln x}{x} dx$$

Proof. We compute

$$\lim_{t\to\infty} \int_1^t \frac{\ln x}{x} dx$$

where

$$\int_1^t \frac{\ln x}{x} dx = \int_0^{\ln t} u du = \frac{\ln^2 t}{2} \to \infty \quad \text{as } t \to \infty$$

□

Problem 3.14. Compute the following improper integral:

$$\int_2^\infty y e^{-3y} dy$$

Proof. We are computing

$$\lim_{s\to\infty} \int_2^s y e^{-3y} dy$$

where

$$\begin{aligned}\int_2^s y e^{-3y} dy &= -\frac{y e^{-3y}}{3}\Big|_2^s + \int_2^s \frac{e^{-3y}}{3} dy \\ &= \frac{-(3s+1)e^{-3s}}{9} + \frac{7}{9}e^{-6} \\ &= \frac{7e^{-6}}{9} \quad \text{as } s \to \infty\end{aligned}$$

□

Definition 3.1 (Midpoint Rule).

$$\int_a^b f(x)\,dx \approx M_n = \Delta x \left[f(\bar{x}_1) + f(\bar{x}_2) + \cdots + f(\bar{x}_n)\right],$$

where

$$\Delta x = \frac{b-a}{n},$$

and

$$\bar{x}_i = \frac{1}{2}(x_{i-1} + x_i) = \text{midpoint of } [x_{i-1}, x_i].$$

Definition 3.2 (Trapezoidal Rule)**.**

$$\int_a^b f(x)\,dx \approx T_n = \frac{\Delta x}{2}\left[f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n)\right],$$

where

$$\Delta x = \frac{b-a}{n}, \qquad x_i = a + i\Delta x.$$

Definition 3.3 (Simpson's Rule)**.**

$$\int_a^b f(x)\,dx \approx S_n = \frac{\Delta x}{3}\big[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)\big],$$

where n is even and

$$\Delta x = \frac{b-a}{n}.$$

Proposition 3.4 (Error Bound for Simpson's Rule)**.** Suppose that

$$|f''(x)| \le K \qquad \text{for } a \le x \le b.$$

If E_T and E_M are the errors in the Trapezoidal and Midpoint Rules, then

$$\boxed{|E_T| \le \frac{K(b-a)^3}{12n^2}} \qquad \text{and} \qquad \boxed{|E_M| \le \frac{K(b-a)^3}{24n^2}}.$$

Now suppose that

$$|f^{(4)}(x)| \le C \qquad \text{for } a \le x \le b.$$

If E_S is the error involved in using Simpson's Rule, then

$$\boxed{|E_S| \le \frac{C(b-a)^5}{180n^4}}.$$

Problem 3.15. Consider the following integral

$$\int_0^1 \cos(x^2)dx$$

Given n, one can compute M_n, T_n. Find n large enough such that M_n or T_n have an error with size no more than 0.0001.

Proof. One can take the derivative to find

$$f'(x) = -2x\sin(x^2), \quad f''(x) = -2\sin(x^2) - 4x^2\cos(x^2)$$

thus $|f''(x)| \le 6$. This gives

$$0.0001 \ge \frac{6}{12n^2} \Rightarrow n \ge \sqrt{5000}$$

for the trapezoidal estimation;

$$0.0001 \ge \frac{6}{24n^2} \Rightarrow n \ge 50$$

□

Chapter 4

Infinite Sequences and Series

Definition 4.1 (limit of sequences). Let $\{a_n\}$ be a sequence, then we say a_n converges to a limit L if given any $\varepsilon > 0$, we can find N large enough such that for $n \geq N$, we have

$$|a_n - L| < \varepsilon$$

Problem 4.1. Does the following sequence have a limit? If so, compute it.

$$a_n = \sin\left(\frac{n\pi}{2}\right)$$

Proof. The sequence looks like

$$\{1, 0, -1, 0, 1, 0, \dots\}$$

so the limit doesn't exist. □

Problem 4.2. Does the following sequence have a limit? If so, compute it.

$$\ln(n+1) - \ln(n)$$

Proof. Choose n large enough,

$$\ln\left(\frac{n+1}{n}\right) = \ln\left(\frac{1+\frac{1}{n}}{1}\right) \to 0$$

as $n \to \infty$. □

Problem 4.3. Does the following sequence have a limit? If so, compute it.

$$a_n = \sqrt{n^2+1} - n$$

Proof. We see

$$\begin{aligned}\sqrt{n^2+1} - n &= \frac{(\sqrt{n^2+1} - n)(\sqrt{n^2+1} + n)}{\sqrt{n^2+1} + n} \\ &= \frac{1}{\sqrt{n^2+1} + n} \to 0\end{aligned}$$

as $n \to \infty$. □

Week 8 (10/26-10/30)

Proposition 4.1. If a series $\sum_{n=1}^{\infty} a_n$ converges, then $a_n \to 0$ as $n \to \infty$.



Warning 4.1. When $a_n \to 0$, it doesn't necessarily imply $\sum_{n=1}^{\infty} a_n < \infty$. Counterexample: $\sum_{n=1}^{\infty} \frac{1}{n}$.

Problem 4.4. Let

$$a_n = \frac{2n}{3n+1}.$$

(a) Determine whether $\{a_n\}$ is convergent.

(b) Determine whether $\displaystyle\sum_{n=1}^{\infty} a_n$ is convergent.

Proof. 1. $\{a_n\}$ as a sequence is convergent, and

$$\lim_{n\to\infty} a_n = \frac{2}{3}$$

2. Since $\lim_{n\to\infty} a_n \neq 0$, the series cannot be convergent.

□

Problem 4.5. Determine whether the series is convergent or divergent. If convergent, find its sum.

$$\sum_{n=1}^{\infty} \frac{e^n}{n^2}$$

Proof. The series does not converge.

$$\lim_{n\to\infty} \frac{e^n}{n^2} = \lim_{n\to\infty} \frac{e^n}{2n} = \lim_{n\to\infty} \frac{e^n}{2} = \infty$$

by L'Hospital. □

Problem 4.6. Determine whether the series is convergent or divergent. If convergent, find its sum.

$$\sum_{n=2}^{\infty} \frac{2}{n^2-1}$$

Proof. The series converges. We can apply the comparison test with $\frac{n^2}{2}$:

$$n^2 - 1 \geq \frac{n^2}{2}$$

for all $n \geq 2$. Thus,

$$\sum_{n=2}^{\infty} \frac{2}{n^2-1} \leq \sum_{n=2}^{\infty} \frac{2}{n^2/2} < \infty$$

Now we compute its sum:

$$\sum_{n=2}^{\infty} \frac{2}{n^2-1} = \sum_{n=2}^{\infty} \frac{1}{n-1} - \frac{1}{n+1}$$

The partial sum $s_k = a_1 + \cdots + a_k$ is then

$$s_k = \left(1 - \frac{1}{3}\right) + \left(\frac{1}{2} - \frac{1}{4}\right) + \cdots + = 1 + \frac{1}{2} - \frac{1}{k} - \frac{1}{k+1}$$

Then letting $k \to \infty$, we see the sum is $3/2$. □

Proposition 4.2. Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms.

(i) If $\sum b_n$ is convergent and $a_n \leq b_n$ for all n, then $\sum a_n$ is also convergent.

(ii) If $\sum b_n$ is divergent and $a_n \geq b_n$ for all n, then $\sum a_n$ is also divergent.

Proposition 4.3. Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms. If

$$\lim_{n\to\infty} \frac{a_n}{b_n} = c,$$

where c is a finite number and $c > 0$, then either both series converge or both diverge.

Problem 4.7. Determine whether the series is convergent or divergent.

$$\sum_{n=1}^{\infty} \frac{1}{n!}$$

Proof. We use the comparison test. For $n \geq 1$, we see

$$2^{n-1} \leq n! \Rightarrow \frac{1}{n!} \leq \frac{1}{2^{n-1}}$$

Thus

$$\sum_{n=1}^{\infty} \frac{1}{n} \leq \sum_{n=1}^{\infty} \frac{1}{2^{n-1}} < \infty$$

as it is a geometric series with ratio $r = \frac{1}{2} < 1$, hence convergent. □

Problem 4.8. Determine whether the series is convergent or divergent.

$$\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$$

Proof. We use the limit comparison test.

$$\lim_{n\to\infty} \frac{\sin\left(\frac{1}{n}\right)}{\frac{1}{n}} = 1$$

Thus applying the limit commparison theorem, we know the series

$$\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right) \text{ and } \sum_{n=1}^{\infty} \frac{1}{n}$$

either both converge or both diverge. Since $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges, we know

$$\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right) \text{ diverges}$$

□

Proposition 4.4. If the alternating series

$$\sum_{n=1}^{\infty} (-1)^{n-1} b_n = b_1 - b_2 + b_3 - b_4 + b_5 - b_6 + \cdots, \qquad b_n > 0$$

satisfies

(i) $b_{n+1} \leq b_n$ for all n,

(ii) $\lim_{n\to\infty} b_n = 0$,

then the series is convergent.

Problem 4.9. Determine whether the series is convergent or divergent.

$$\sum_{n=1}^{\infty} (-1)^{n+1} n e^{-n}$$

Proof. We'd like to use the alternating series test. We need to check

$$b_n = \frac{n}{e^n}$$

satisfy

1. $\lim_{n\to\infty} b_n = 0$.
2. $b_{n+1} \leq b_n$ for all n.

For the first check:

$$\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{n}{e^n} = \lim_{n\to\infty} \frac{1}{e^n} = 0$$

by L'Hospital. For the second check,

$$\frac{b_{n+1}}{b_n} = \frac{(n+1)e^n}{e^{n+1}n} = \frac{n+1}{n} \cdot \frac{1}{e} \leq 1$$

for all n. Thus the alternating series test applies, hence the series is convergent. □

Week 9 (11/2-11/6)

Definition 4.2. A power series centered at $x = 0$ is an infinite series $\sum_{n=0}^{\infty} c_n x^n$, where c_n are constant coefficients. A power series centered at $x = a$ is of the form $\sum_{n=0}^{\infty} c_n (x-a)^n$.

You can think of the power series as a function who is only defined in some regions. (For a function to make sense, we need to assign a finite value to every point of x, but that is not necessarily the case for power series, hence limited domain).

Example 4.1. If $|x| < 1$, the geometric series

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

where the LHS is a power series centered at 0, with $c_n = 1$.

$$f(x) = 1 + x + x^2 + x^3 + \dots$$

Why do we care about power series? Next topic: Taylor series. Sometimes we can write a complicated function with a power series. One nice thing about power series is that we can differentiate/integrate more easily!

Example 4.2. You will learn to compute this later:

$$e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n$$

for all $x \in \mathbb{R}$.

Example 4.3.

$$\sin(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^6}{6!}$$

for all $x \in \mathbb{R}$. For x close to 0, we know intuitively why

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

Are there functions that don't have a power series expansion?

Example 4.4. A smooth function that doesn't have a power series expansion:

$$f(x) = \begin{cases} e^{-\frac{1}{x^2}}, x \geq 0 \\ 0, x < 0 \end{cases}$$

Proposition 4.5. Given a power series $\sum_{n=0}^{\infty} c_n (x-a)^n$, one of three following must happen:

1. The series converges for all $x \in \mathbb{R}$.
2. The series only converges for $x = a$.
3. There exists an $R > 0$ such that whenever $|x - a| < R$, the series converges; when $|x - a| > R$, the series diverges; when $|x - a| = R$, we don't know.

This radius of convergence is closely related to series convergence and all the tests introduced before such as ratio and root tests.

Example 4.5.

$$\sum_{n=0}^{\infty} n!x^n$$

The radius of convergence is 0.

Problem 4.10. Find the radius of convergence for the following power series

$$\sum_{n=1}^{\infty} n^n x^n$$

Proof.

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}(n+1)\left(1+\frac{1}{n}\right)^n|x| = \infty$$

for any $x \neq 0$. Hence the radius of convergence is 0. □

Problem 4.11. Find for which x the following power series converge.

$$\sum_{n=1}^{\infty} \frac{x^{2n}}{n!}$$

Proof. For $x \neq 0$,

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{x^2}{n+1} = 0$$

for any fixed $x \neq 0$, this converges. It also converges at $x = 0$. Thus the power series converge for all $x \in \mathbb{R}$. □

Problem 4.12. If k is a positive integer, find the radius of convergence of the series

$$\sum_{n=0}^{\infty} \frac{(n!)^k}{(kn)!}x^n.$$

Proof.

$$\lim_{n\to\infty}\frac{(n+1)!^k(kn)!}{(kn+k)!(n!)^k}|x| = \lim_{n\to\infty}\frac{(n+1)^k}{(kn+1)\dots(kn+k)}|x| = \frac{1}{k^k}|x|$$

which means the radius of convergence is k^k.

□

Week 10 (11/9-11/13)

Example 4.6. The following power series can be rewritten as follows:

$$\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n$$

whenever $|x| < 1$. Similarly,

$$\frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n}$$

whenever $|x| < 1$.

Problem 4.13. Find the power series expansion and the convergence radius for

$$\frac{x}{2x^2+1}$$

Proof.

$$\sum_{n=0}^{\infty} (-1)^n 2^n x^{2n+1}$$

whenever $|2x^2| < 1 \Rightarrow |x| < 1/\sqrt{2}$. □

Problem 4.14. Find the power series of $\ln(1+x)$ and the radius of convergence.

Proof. We first note the power series of the derivative $\frac{1}{1+x}$ in $|x| < 1$ is

$$\sum_{n=0}^{\infty} (-1)^n x^n = 1 - x + x^2 - x^3 + \dots$$

Hence

$$\ln(1+x) = C + x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} = C + \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}$$

when $x = 0$, we find out $C = 0$. Hence

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}$$

whenever $|x| < 1$. □

Problem 4.15. Use differentiation to find a power series representation for

$$f(x) = \frac{1}{(1+x)^2}.$$

What is the radius of convergence?

Proof.

$$\frac{1}{(1+x)^2} = \sum_{n=1}^{\infty}(-1)^{n+1}nx^{n-1} = \sum_{n=0}^{\infty}(-1)^n(n+1)x^n$$

when $|x| < 1$. □

Proposition 4.6 (Taylor series is the expansion). If f has a power series representation (expansion) at a, that is, if

$$f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n, \qquad |x-a| < R,$$

then its coefficients are given by the formula

$$c_n = \frac{f^{(n)}(a)}{n!}.$$

In other words,

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots$$

Proposition 4.7 (Taylor's inequality). If $|f^{(n+1)}(x)| \le M$ for $|x-a| \le d$, then the remainder $R_n(x)$ of the Taylor series satisfies the inequality

$$|R_n(x)| \le \frac{M}{(n+1)!}|x-a|^{n+1} \qquad \text{for } |x-a| \le d.$$

Problem 4.16. Let $f(x) = \frac{1}{1+x}$. Find its 4th order Taylor polynomial about $x = 2$.

Proof. Another way to do this is since we know

$$\frac{1}{1+x} = \sum_{n=0}^{\infty}(-1)^n x^n$$

for $|x| < 1$. Now we want it to be centered at 2, we rewrite

$$\frac{1}{1+x} = \frac{1}{3}\cdot\frac{1}{1+\frac{x-2}{3}} = \sum_{n=0}^{\infty}\frac{(-1)^n}{3^{n+1}}(x-2)^n$$

where $|x-2| < 3$. □

Problem 4.17. Let $f(x) = x\cos x$. Find its Taylor series about $x = 0$ and its radius of convergence.

Proof. We know

$$\cos x = \sum_{n=0}^{\infty}(-1)^n\frac{x^{2n}}{(2n)!}$$

Thus the Taylor series is

$$\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n+1}}{(2n)!}$$

And we can use the ratio test to see the radius of convergence is $R = \infty$. □

Problem 4.18. For the following series

$$\sum_{n=0}^{\infty} \frac{3^n}{5^n n!}$$

Find its sum.

Proof. We recall

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

Thus the sum is $e^{\frac{3}{5}}$. □